Beyond the right recoil saddle: a two-branch Barnes–Bessel diagnostic

Alan Day Chengheng*

Working research note — 30 October 2026

Abstract

Exact negative-side reflection and a compact-kernel limit lead to an auxiliary contour with two modified-Bessel contributions. Finite-parameter quadrature shows a large additional contribution for some imaginary energy transfers. This revises the prospective global single-saddle strategy and preserves the preceding local result. Neither a full Barnes-to-Bessel limit nor a joint growing-mass theorem is proved here. Numerical errors quoted are empirical diagnostics.

1 Object and exact negative reflection

The de Sitter radius is one. We retain the internal chiral graph amplitude obtained in the preceding fixed-mass $\kappa \to \infty$ limit, then study its masses. No physical state, observable algebra, external insertion or energy cycle is newly identified here. This is not a finite-G OTOC. For $u = v = 0$, $K = \nu^2/(2M) > 0$, $\omega = iKy$, set

$$a_j = M + \nu/2 \ (j = 1, \dots, 4), \qquad r = (M + \nu + \omega/2, 2M, M + \nu - \omega/2).$$

The exact kernel and chiral integral are

$$B(z) = e^{-i\pi z} \frac{\prod_a \Gamma(z + ia) \prod_r \Gamma(\frac{1}{2} - z - ir)}{\Gamma(z + \frac{1}{2})}, \qquad \mathfrak{B} = \frac{1}{2\pi i} \int_{\mathbb{R} - ic} B(z) \, \mathrm{d}z.$$

For $w = -z$, Euler reflection [1] gives

$$B(-w) = H_-(w) E_-(w), \tag{1}$$

$$H_-(w) = 8\pi^3 e^{4\pi i w + \pi \sum r} \frac{\Gamma(w + \frac{1}{2}) \prod_r \Gamma(w + \frac{1}{2} - ir)}{\prod_a \Gamma(1 + w - ia)}, \tag{2}$$

$$E_-(w) = \frac{1 + e^{2\pi i w}}{\prod_a (1 - e^{2\pi i (w - ia)})}. \tag{3}$$

No periodic denominator is discarded. This reflection identity also holds for the general real transverse labels. With $f(t) = t(\log t - 1)$, define

$$\Phi(z) = 2\pi i z - \pi \sum r + \sum_a f(z + ia) - \sum_r f(z + ir) - f(z).$$

*AI research assistant (OpenAI).

2 Phase comparison and a computable contour

For real a, conjugation-invariant r, and compatible right-half-plane logarithms, the negative phase obeys

$$\Psi(w) = 4\pi i w + \pi \sum r + f(w) + \sum_r f(w - ir) - \sum_a f(w - ia) = 2\pi i w - \overline{\Phi(\bar{w})}.$$

Let $z_+ = x_+ - ic_+$ be the right saddle. Write $\prod(s-a) - s\prod(s-r) = A_q s^2 + bs + c_0$; then $s_+ = (-b + i\sqrt{-\Delta})/(2A_q)$, $z_+ = -is_+$, $\Delta < 0$. The difference of real phase values at conjugate points is

$$D = \Re\Phi(z_+) - \Re\Psi(\bar{z}_+) = 2\pi c_+ + 2\Re\Phi(z_+).$$

This does not determine a contour coefficient. For $\epsilon = \nu/M$, $\delta = M + \nu - c_+$, the symmetric small-ϵ expansion at bounded y is

$$D/K = 2F(y) - 2\pi\delta/K + O(\epsilon^2), \qquad F(y) = \sqrt{1+y^2} - y \operatorname{arsinh} y.$$

Choose $L = x_+ > K|y|/2$, $c = M + \nu - d$ with $\max a < c < \min \Re r$. The negative tail may be deformed downward at $x = -L$. The swept lower-left region has no poles: the $\Gamma(z + ia)$ poles lie at height $-a > -c$, and the other pole rows lie to the right of $-L$. Reflection and Stirling remove distant arcs at fixed parameters. The orientation is upward on the left ray:

$$\int_{\mathbb{R}-ic} B(z)\, \mathrm{d}z = \int_{-L}^{\infty} B(x - ic)\, \mathrm{d}x + i \int_0^{\infty} B(-L - i(c+t))\, \mathrm{d}t.$$

This fixed-parameter deformation supplies no mass-uniform tail bound.

3 Compact-kernel limit

Put $t = z + i(M+\nu)$, $\alpha = Ky$, and

$$C_{M,\nu} = e^{-\pi(M+\nu)} \frac{\Gamma(-i\nu/2)^4 \Gamma(\frac{1}{2} - i(M-\nu))}{\Gamma(\frac{1}{2} - i(M+\nu))}.$$

At fixed K, α, $M = \nu^2/(2K)$, $\nu \to \infty$, Gamma-ratio asymptotics [2] give locally away from poles

$$\frac{B(t - i(M+\nu))}{C_{M,\nu}} \longrightarrow U_{K,\alpha}(t), \tag{4}$$

$$U_{K,\alpha}(t) = e^{-2\pi i t} (K/2)^{2t} \Gamma(\tfrac{1}{2} - t + \alpha/2) \Gamma(\tfrac{1}{2} - t - \alpha/2). \tag{5}$$

The leading powers are $(-i\nu/2)^{4t}[-i(M-\nu)]^{-t}[-i(M+\nu)]^{-t} e^{-i\pi t}$; on these branches they tend to $e^{-2\pi i t}(K/2)^{2t}$. This is compact convergence, not an exchanged full integral limit.

4 Exact auxiliary Bessel integral

For real α, $K > 0$, choose $\mathcal{C}$ from $d_0 - i\infty$ to $d_0 + i\eta$, then right to $+\infty + i\eta$, where $d_0 < \frac{1}{2} - |\alpha|/2$, $\eta > 0$. Set $J = (2\pi i)^{-1} \int_{\mathcal{C}} U \, \mathrm{d}t$. Its downward tail is a power times $e^{-3\pi|\Im t|}$; reflection on the right tail gives reciprocal Gamma factors and factorial decay. Close clockwise below the horizontal segment, with boundaries avoiding the real poles.

For noninteger α, summing the two pole families $t = \frac{1}{2} \pm \alpha/2 + n$ gives

$$\begin{aligned} J = & -e^{-i\pi\alpha}\Gamma(-\alpha)(K/2)^{1+\alpha}{}_0F_1(1+\alpha; K^2/4) \\ & -e^{i\pi\alpha}\Gamma(\alpha)(K/2)^{1-\alpha}{}_0F_1(1-\alpha; K^2/4). \end{aligned} \tag{6}$$

The residue sign includes both the derivative of the Gamma argument and the clockwise closure. Using the I series and the standard $I_{-\alpha} - I_\alpha$ connection [3, 4],

$$J = \frac{\pi K}{2\sin(\pi\alpha)} \left(e^{-i\pi\alpha} I_\alpha(K) - e^{i\pi\alpha} I_{-\alpha}(K) \right) \tag{7}$$

$$= -K \left[e^{i\pi\alpha} K_\alpha(K) + i\pi I_\alpha(K) \right] = -K K_\alpha(K e^{-i\pi}). \tag{8}$$

The last expression denotes continuation below the negative axis [5]. At integer order continue the *complete sum*; the separate residue coefficients are singular. This is an exact auxiliary identity, not yet the full Barnes limit, and not a claim of a new special-function identity.

Two exponential scales

For fixed $y > 0$, $K \to \infty$, large-order expansions [6] give

$$I_{Ky}(K) \sim \frac{e^{KF(y)}}{\sqrt{2\pi K\sqrt{1+y^2}}}, \qquad K_{Ky}(K) \sim \sqrt{\frac{\pi}{2K\sqrt{1+y^2}}} e^{-KF(y)}.$$

Thus $K_{Ky}(K)/(\pi I_{Ky}(K)) \sim e^{-2KF(y)}$. Since $F'(y) = -\operatorname{arsinh} y < 0$, its positive zero is unique:

$$y_c \simeq 1.5088795615383199289.$$

This is an auxiliary exponential-magnitude crossover, not a proved physical critical point or a Stokes transition of the full graph. Phase cancellation of the complete sum must still be controlled.

5 A first correction and finite-parameter evidence

At fixed K, α, t, the compact Gamma expansion gives

$$B/C_{M,\nu} = U_{K,\alpha}(t)\left[1 + \frac{4i}{\nu}(t^2 - t) + O_{K,\alpha,t}(\nu^{-2})\right].$$

The four large arguments $-i\nu/2$ supply $4t(t-1)/(2(-i\nu/2)) = 4i(t^2-t)/\nu$; the remaining arguments and power corrections enter at order ν^{-2}. Let $\mathcal{D} = (K/2)\partial_K$, *holding* α *fixed.* Then $\mathcal{D}U = tU$, and the Bessel equation yields exactly

$$(\mathcal{D}^2 - \mathcal{D})J = (K^2 + \alpha^2 - 1)J/4.$$

Formal integration therefore predicts the normalized first-order phase $e^{i\alpha^2/\nu}$. This exponential completes a first-order term; it is not a derived all-orders resummation or an integrated error theorem.

For $u = v = 0$, $\nu = 4K^3$, $M = \nu^2/(2K)$, define

$$g_+ = 8\pi^3 \prod_a (z_+ + ia)^{-1/2}, \qquad \mathfrak{B}_{\mathrm{loc,lead}} = \frac{g_+ e^{\Phi_+}}{i\sqrt{2\pi(-\Phi_+'')}}.$$

The numerical full-contour to local-leading ratios are

K	y	$\mathfrak{B}/\mathfrak{B}_{\mathrm{loc,lead}}$ (approximate)
4	0	$1.0377303 - 0.0016775i$
4	2	$0.7574552 - 182.4665112i$
8	0	$1.0169041 - 0.0001656i$
8	1	$1.0014509 - 0.0003609i$
8	2	$-4.4614748 - 33541.8312194i$
16	0	$1.0081070 - 0.0000205i$
16	2	$-22852.0717 - 1.1232773362\,10^9 i$

These finite values motivate a second contribution; they do not prove an asymptotic counterexample along the entire sequence. For $R = \mathfrak{B}(K, Ky)/\mathfrak{B}(K, 0)$, $R_J = J(K, Ky)/J(K, 0)$, we find

K	y	$\lvert R/R_J - 1\rvert$	$\lvert R/(R_J e^{i(Ky)^2/\nu}) - 1\rvert$
4	2	0.24889358	0.00389292
8	1	0.03124995	0.00007430
8	2	0.12488755	0.00049349
16	2	0.06248789	0.00006189

There is no fitted phase parameter. Its integrated validity is open.

6 Controls, limitations, and next steps

We use 20/36 Gauss nodes per segment, 50/65 decimal digits, $d = 1$, $X = \max(32, 8K)$, and downward extent $T = 18$. For $y = 2$, additional runs use $d = 0.75$, $X = \max(32, 10K)$, $T = 24$, and 90 digits. The segment cancellation ratio reaches about 60.5; all relative comparisons use the complete complex sum. Extent changes test truncation empirically, without certifying the omitted infinite tails.

All 34 checks pass: 12 reflection/phase identities, 6 auxiliary Bessel/differential identities, 7 quadrature refinements, and 9 contour/extent/precision comparisons. Parameters, segment values, refinement changes, Python and mpmath versions, and code/input hashes are stored in `results/stage4/recoil-global/diagnostics.json`. The executable is `code/recoil_global_probe.py`.

The preceding local result is retained. The next objective is a *two-branch* analysis rather than a presupposed right-saddle dominance for all bounded imaginary transfers. First prove a full fixed-K, α Barnes-to-Bessel limit. The unresolved requirement is a mass-uniform integrable tail bound, especially on the positive horizontal segment. A useful partition is $x \ll \nu$, $\nu \lesssim x \lesssim M$, and $x \gtrsim M$, with overlap estimates and all exact periodic factors retained. Compact convergence cannot replace this bound.

Only afterwards let K grow, specify its relation to ν, and establish integrated correction estimates. Near comparable branch magnitudes, control absolute errors before dividing by a potentially small sum. Transverse u, v directions and exterior tails require separate estimates.

The external energy cycles, insertion normalization, disconnected baseline, and microscopic dictionary remain open. Taking the fixed-mass κ limit before mass growth does not establish a joint finite-G limit. Complex energies require analytic continuation of the chiral factors, not a newly imposed modulus square. This is an author self-reviewed research draft.

References

[1] NIST DLMF, §5.5, functional relations, https://dlmf.nist.gov/5.5.

[2] NIST DLMF, §5.11, asymptotic expansions, https://dlmf.nist.gov/5.11.

[3] NIST DLMF, §10.25, modified Bessel definitions, https://dlmf.nist.gov/10.25.

[4] NIST DLMF, §10.27, connection formulas, https://dlmf.nist.gov/10.27.

[5] NIST DLMF, §10.34, especially 10.34.2, https://dlmf.nist.gov/10.34.

[6] NIST DLMF, §10.41(ii), large-order expansions, https://dlmf.nist.gov/10.41.